

Proximity Sensor

Objective:

To detect the presence and relative distance of nearby objects using the built-in APDS9960 proximity sensor on the Arduino Nano 33 BLE Sense.

Procedure:

1. Opened Arduino IDE.
2. Installed the Arduino_APDS9960 library.
3. Opened the ProximitySensor example from:
4. File → Examples → Arduino_APDS9960 → ProximitySensor
5. Uploaded the program to the Arduino Nano 33 BLE Sense board.
6. Opened the Serial Monitor at 9600 baud rate.
7. Moved my hand towards and away from the sensor.
8. Observed the proximity values displayed in the Serial Monitor.

Working Principle:

The APDS9960 proximity sensor uses:

- Infrared (IR) LED
- Infrared Receiver

The sensor emits infrared light and measures the reflected light from nearby objects.

The output is a raw proximity value:

0 → Object very close

255 → Object far away

The value changes according to the distance between the object and the sensor.

Observations:

- The proximity value changed when an object was moved near the sensor.
- Different distances produced different proximity readings.
- The sensor successfully detected the presence of nearby objects.
- The readings varied continuously as the object moved.



0

35

70

68

74